

A diskcyclic operator whose repeated direct sums are not bitransitive

Solution packet for arXiv:1507.01166

11 August 2026

Candidate full negative answer. There is a diskcyclic operator T on a separable complex Banach space such that $T \oplus T$ is not 2-bitransitive. In fact the k -fold direct sum of this same T is not k -bitransitive for any $k \geq 2$.

1 The question and definitions

For $S \in \mathcal{B}(X)$, a vector $x \in X$ is *diskcyclic* when

$$\{\alpha S^n x : n \geq 0, |\alpha| \leq 1\}$$

is dense in X . The direct sum $S_1 \oplus \cdots \oplus S_k$ is *k-bitransitive* if, for all nonempty open $U_i, V_i \subset X$, there are a common $n \geq 0$ and nonzero $\alpha_i \in \mathbb{D}$ such that

$$S_i^n(\alpha_i U_i) \cap V_i \neq \emptyset \quad (1 \leq i \leq k).$$

Bamerni, Kılıçman and Noorani prove positive results under common diskcyclic-criterion or compoundness assumptions, and end with the question shown below.

that the 1 remains open. Therefore, one may answer a special case of that question, that is, when all k diskcyclic operators are identical. Particularly.

If T is a diskcyclic operator. What about the k -fold direct sum of T ?

References

- [1] E. Akin, E. Glasner, W. Huang, S. Shao, and X. Ye, Sufficient conditions under which a transitive system is chaotic, *Israel Journal of Mathematics* 29 (2018)

Figure 1: The exact final question on page 10 of the official arXiv PDF.

2 Statement of the answer

Theorem 1. *For every $\lambda > 1$ there are a separable complex Banach space X and $T \in \mathcal{B}(X)$ such that:*

1. T is diskcyclic;
2. $T^{\oplus k}$ is not k -bitransitive for any $k \geq 2$.

The only existence input is the theorem of de la Rosa and Read [2]: there are a separable complex Banach space Y and a hypercyclic $R \in \mathcal{B}(Y)$ for which $R \oplus R$ is not hypercyclic. By the Birkhoff transitivity theorem, $R \oplus R$ is therefore not topologically transitive.

3 Idea of the proof

Put a fixed one-dimensional coordinate beside R , then amplify both coordinates at the same exponential rate:

$$X = Y \oplus \mathbb{C}, \quad T = \lambda(R \oplus I_{\mathbb{C}}).$$

The factor λ^n makes $(x, 1)$ diskcyclic whenever x is hypercyclic for R : at time n the disk scalar can implement every effective scalar of modulus at most λ^n .

The fixed coordinate also detects those effective scalars. If an input and its image both have scalar coordinate close to 1, their quotient is close to 1. Thus alleged 2-bitransitivity of $T \oplus T$, tested on such scalar-coordinate neighborhoods, would remove the two independent disk scalars and force ordinary transitivity of $R \oplus R$. This contradicts the choice of R .

4 Proof

Choose a hypercyclic vector x for R . We first show that $(x, 1)$ is diskcyclic for T . Since

$$\alpha T^n(x, 1) = \beta(R^n x, 1), \quad \beta = \alpha \lambda^n,$$

the available effective scalars at time n are exactly $|\beta| \leq \lambda^n$.

Let $(y, c) \in Y \oplus \mathbb{C}$ and $\varepsilon > 0$. If $c \neq 0$, tail-density of the hypercyclic orbit lets us choose arbitrarily large n with

$$\|R^n x - y/c\| < \varepsilon/|c| \quad \text{and} \quad \lambda^n \geq |c|.$$

Taking $\beta = c$ gives the required approximation. If $c = 0$, choose $0 < |\beta| < \varepsilon$, then choose arbitrarily large n such that

$$\|R^n x - y/\beta\| < \varepsilon/|\beta| \quad \text{and} \quad \lambda^n \geq |\beta|.$$

Now $\beta(R^n x, 1)$ is within 2ε of $(y, 0)$ in the sum norm. In both cases $\alpha = \beta/\lambda^n \in \mathbb{D}$, so T is diskcyclic.

Suppose, toward a contradiction, that $T \oplus T$ is 2-bitransitive. Let $U_i, V_i \subset Y$ be arbitrary nonempty open sets, $i = 1, 2$. Choose $v_i \in V_i$. Continuity of $(\gamma, w) \mapsto \gamma^{-1}w$ at $(1, v_i)$ gives nonempty open $W_i \subset V_i$ and a common $\eta > 0$ such that

$$|\gamma - 1| < \eta, w \in W_i \implies \gamma^{-1}w \in V_i. \quad (1)$$

Choose $0 < \delta < 1$ so small that

$$\frac{2\delta}{1-\delta} < \eta.$$

Apply 2-bitransitivity to the two input/output pairs

$$A_i = U_i \times D(1, \delta), \quad B_i = W_i \times D(1, \delta).$$

There are a common n , nonzero $\alpha_i \in \mathbb{D}$, and $(u_i, z_i) \in A_i$ such that

$$\alpha_i T^n(u_i, z_i) = (\beta_i R^n u_i, \beta_i z_i) \in B_i, \quad \beta_i = \alpha_i \lambda^n.$$

Both z_i and $\beta_i z_i$ belong to $D(1, \delta)$, whence

$$|\beta_i - 1| = \frac{|\beta_i z_i - z_i|}{|z_i|} < \frac{2\delta}{1-\delta} < \eta. \quad (2)$$

Moreover $\beta_i R^n u_i \in W_i$. Equations (1)–(2) therefore give $R^n u_i \in V_i$, while $u_i \in U_i$, for both i at the same n . Thus

$$(R \oplus R)^n(U_1 \times U_2) \cap (V_1 \times V_2) \neq \emptyset.$$

Since the four open sets were arbitrary, $R \oplus R$ is topologically transitive, the required contradiction.

Finally, if $T^{\oplus k}$ were k -bitransitive for some $k > 2$, take the remaining $k - 2$ input and output open sets to be all of X . The first two coordinates would make $T \oplus T$ 2-bitransitive. Hence no $k \geq 2$ works, proving Theorem 1.

5 Verification, scope, and novelty

The proof uses no computation. The two limiting issues in the construction are explicit: $\lambda > 1$ supplies arbitrarily large effective disk scalars, while $\delta < 1$ keeps the pinning denominator $|z_i|$ away from zero. Independent scalars are allowed throughout; the added fixed coordinate is precisely what pins each one separately.

The run registry, parsed corpus, exact question phrase, exact title, and bounded web searches through 11 August 2026 were checked for “diskcyclic k -fold direct sum”, “diskcyclic direct sum k -bitransitive”, and citations/close variants of arXiv:1507.01166. We found the source paper and its sufficient criteria, but no later direct answer to its final question. The de la Rosa–Read paper supplies the hypercyclic operator used here but does not state this diskcyclic construction. Novelty confidence is moderate.

Human review should verify only the interpretation of the source’s final question as asking whether the identical direct sum is k -bitransitive, and the standard hypercyclicity–transitivity equivalence. Conditional on that intended interpretation, the theorem is a complete negative answer.

References

- [1] N. Bamerni, A. Kılıçman, and M. S. M. Noorani, *k -bitransitive and compound operators on Banach spaces*, arXiv:1507.01166 (2015).
- [2] M. de la Rosa and C. Read, *A hypercyclic operator whose direct sum $T \oplus T$ is not hypercyclic*, J. Operator Theory **61** (2009), 369–380.